# E-Commerce Product & User Behavior Analytics

Project Analysis Report

A portfolio project demonstrating data cleaning, sales analysis, customer behaviour, conversion funnel metrics, and an introductory A/B testing dashboard.

| | |
|---|---|
| **Tools** | Microsoft Excel, Python, SQL, Power BI |
| **Data** | Synthetic e-commerce data created for this project |
| **Project Scope** | Users, events, products, orders, and experiment assignments |

Important: This report uses simulated data for learning and portfolio demonstration. The findings should not be presented as results from a real business.

# 1. Executive Summary

This project analyses a simulated e-commerce business to demonstrate how spreadsheet, SQL, Python, and business intelligence tools can be used to explore product performance, customer purchasing, and user activity.

The current Excel analysis reports 10,000 registered users, 2,884 order records, total revenue of ■5,78,46,205, and 5,717 units sold. Of the registered users, 2,545 placed at least one order. The repeat-customer calculation identified 315 customers with more than one order, giving a repeat purchase rate of approximately 12.38% among purchasing customers.

The Power BI A/B testing page displays a Control conversion rate of 25.67% and a Treatment conversion rate of 25.55%. This is a difference of -0.12 percentage points based on the displayed rounded rates. The displayed rates alone are not enough to conclude whether the difference is statistically significant.

# 2. Project Objectives

• Summarise overall orders, revenue, and quantity sold.

• Check selected dataset identifiers for missing values and duplicates.

• Explore product-level and category-level revenue.

• Measure customer purchasing and repeat-purchase behaviour.

• Summarise user counts at product-view, add-to-cart, checkout, and purchase events.

• Present experiment-group conversion rates and communicate limitations responsibly.

# 3. Tools and Dataset

**Tools:** Microsoft Excel for formulas and charts; Python and its data-analysis libraries for data preparation and analysis; SQL for querying; and Power BI for interactive dashboards.

**Datasets:** users.csv, events.csv, products.csv, orders.csv, and experiments.csv. The files represent registered users, user events, product attributes, order records, and experiment assignments. All data is synthetic.

The Excel workbook includes data-quality checks, product analysis, funnel analysis, customer analysis, and a summary dashboard. The Power BI project contains pages for product overview, conversion funnel, retention/cohort analysis, A/B testing, customer and revenue analytics, and product recommendations.

# 4. Data Quality Checks

The initial Excel checks returned zero missing values in the selected identifier columns for Users, Events, Products, Orders, and Experiments. Duplicate-ID checks returned zero for the identifiers checked, including event_id and user_id in the experiment table.

The experiment_id value was repeated across records because it identifies the experiment itself (checkout_redesign_01), not each individual assignment. Therefore, it should not be treated as a unique row identifier.

**Limitation:** These checks only cover selected ID columns. They do not prove that every field is complete, valid, or logically consistent. A full quality review should also examine dates, categories, quantities, amounts, relationships between tables, and repeated assignments.

# 5. Sales and Product Analysis

| Metric | Result |
|---|---|
| Order records | 2,884 |
| Quantity sold | 5,717 |
| Total revenue | ■5,78,46,205.00 |

| Metric | Result |
|---|---|
| Average order value | ■20,049.31 |
| Products in product table | 50 |
| Best-selling product by quantity | Product 030 |
| Highest revenue for a single product | ■23,83,944.00 |

Average order value is calculated as total revenue divided by order records. Product 030 was identified as the best-selling product by quantity in the current worksheet. Category-level findings should be interpreted after checking that revenue is correctly attributed through product IDs.

## 6. Customer Analysis

The workbook reports 10,000 registered users and 2,545 purchasing customers. The customer purchase rate is 2,545 ÷ 10,000 = 25.45%.

The analysis identifies 315 repeat customers. Dividing 315 by 2,545 purchasing customers gives a repeat purchase rate of approximately 12.38%. The remaining 87.62% of purchasing customers placed only one order in the dataset, according to the current repeat-customer definition.

Average revenue per purchasing customer is reported as ■22,729.35. This is total order revenue divided by the number of unique purchasing customers; it is not the same as average order value.

## 7. Conversion Funnel

The Excel funnel currently reports the following distinct-user counts:

| Event stage | Unique users | Ratio to previous stage |
|---|---|---|
| Product views | 10,000 | 100.00% (baseline) |
| Add to cart | 6,765 | 67.65% of product-view users |
| Checkout | 4,786 | 70.75% of add-to-cart users |
| Purchases | 2,545 | 53.18% of checkout users |

These are ratios of aggregate unique-user counts at each event type. They do not confirm that the same users completed each stage in the correct chronological order. A production-grade funnel should link sessions or users and verify event sequence and time windows. The current event name for checkout should also be checked for spelling consistency (for example, begin_checkout versus beign_checkout).

## 8. A/B Testing

The Power BI page displays the following experiment-group rates:

| Experiment group | Displayed conversion rate |
|---|---|
| Control | 25.67% |
| Treatment | 25.55% |
| Treatment minus Control | -0.12 percentage points* |

*Difference calculated from the displayed, rounded rates. Before making a business recommendation, verify that the converted field accurately represents the chosen conversion definition, check the group sizes and conversion counts, and calculate a confidence interval or statistical test. The rates by themselves do not establish a winning variant.

## 9. Business Recommendations

• **Investigate repeat purchases:** Review customer journeys and product categories to understand why most purchasing customers have only one order. Consider relevant follow-up offers or replenishment reminders where appropriate.

• **Improve funnel measurement:** Validate event naming and track ordered transitions by user or session before interpreting stage-to-stage conversion as true funnel conversion.

• **Use experiment guardrails:** Confirm assignment integrity, sample sizes, conversion definitions, and statistical uncertainty before rolling out the checkout redesign.

• **Review product performance:** Compare revenue, quantity, margin, and return rates rather than relying on revenue alone.

• **Segment responsibly:** Compare channels and device types with consistent denominators and enough observations to avoid over-interpreting small groups.

## 10. Limitations and Next Steps

The data is simulated and cannot represent actual customer behaviour. The current funnel is based on aggregate event counts and does not prove sequential movement. The A/B test page displays rates, but statistical significance has not been established in this report. Some measures and relationships should be validated against the raw CSV files before the results are used for decision-making.

Recommended next steps are to verify the exact event-name values, validate revenue joins by product ID, check data types and referential integrity, add a true cohort-retention calculation, and calculate an appropriate A/B test confidence interval or hypothesis test from the underlying counts.

## 11. Conclusion

This project demonstrates an end-to-end analytics workflow using spreadsheet analysis and BI reporting. It covers data-quality checks, product revenue, customer purchasing behaviour, conversion event counts, and experiment-group comparison. Its strongest portfolio value is the documented workflow and transparent communication of assumptions and limitations—not a claim that the simulated figures describe a real company.